

Kevin Chang

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Education

University of California, Los Angeles (UCLA)

Expected grad: June 2028

B.S. in Electrical Engineering | GPA: 3.9 | Dean's Honor List | Eta Kappa Nu (ECE Honor Society)

Distinctions

MakeMIT x Harvard - 2nd Place

2026

Won 2nd place at MakeMIT x Harvard's entertainment track (6 tracks total), the largest hardware hackathon in the US at the time with 400+ participants. Collaborated with a team to engineer a high-speed volumetric display that generated 3D voxel-based visuals using persistence of vision by rotating an LED matrix at several hundred RPM.

Github x Hack Club Hardware Hackathon - 1st Place

2025

Won 1st out of 160 participants at Undercity, the world's largest teen hardware hackathon hosted by Github and Hack Club. Built an automatic toilet paper folder that dispenses and folds toilet paper using 2 stepper motors.

Other Awards: Scholastic Art and Writing Awards Gold Key, Presidential Volunteer Service Award Teen Gold, Chinese American Citizens Alliance National Essay Contest Top 10

Publications

Chang, Kevin. "Tactile Web Browser: A Modular 2D Tactile Interface for Spatial Web Navigation."

Published in *IEEE Xplore* and presented at *IEEE MWSCAS 2026 Student Design Projects*

- Developed a modular 2D tactile interface that converts webpage layouts into physical spatial representations, enabling blind users to perceive webpage structure beyond the 1D limitations of traditional screen readers. Reduced cost by 37 times, from existing \$15k+ multi-line refreshable Braille displays to ~\$400.
- Designed scalable 3×6 bistable-solenoid modules forming 12×6 and other custom tactile arrays, distinct raised keycap configurations (Tetris-like shapes) encode different HTML elements (links, buttons, headers) for rapid spatial recognition..
- Engineered a distributed embedded architecture integrating TB6612FNG drivers, 74HC595 SPI expanders, keyboard matrices, Arduino Nano controllers, I2C communication, and a protected 180W power system.
- Developed a web-to-tactile rendering pipeline to convert webpage structures into real-time actuator commands.

Experience

Merazine - PCB Design Contractor

2026 March - July

Designed and iterated a production-ready ESP32-S3 consumer electronics PCB for a fashion e-commerce startup (250k+ users, \$1M raised), integrating USB-C power/data, Li-ion battery management, Wi-Fi/BLE connectivity, I2S audio, MEMS microphone input, SPI flash/SD storage, debug/test features, and manufacturability considerations while validating hardware bring-up, peripheral integration, and prototype testing for a 100-unit pilot manufacturing run. Validated end-to-end embedded audio and connectivity pipelines, including SD card storage, I2S microphone/speaker subsystems, Wi-Fi communication, BLE provisioning, and ESP32-S3 firmware integration for cloud-based voice processing workflows.

UCLA Emergence of Communication Lab - PCB Design Intern

2026 Jan - June

Collaborated with a team of 4 to design NSF-funded wearable infant sensing prototypes capturing synchronized audiovisual data to study early speech development. Helped develop a 4-layer PCB and debug ESP-IDF firmware integrating ESP32-S3 with OV2640 camera, ICS-43434 mics, OLED display, and transduction microphones for synchronized real-time audiovisual logging to SD over 4-bit SDMMC, including battery-safe buck/boost power design and RTOS-based peripheral coordination.

UCSB - Researcher

2023 June - Aug

Participated in Summer Research Academies and gained lab experience in neuroscience. Co-authored paper: "The Role of the Hippocampus in Context-dependent Decision Making and Continual Learning."

International Youth Neuroscience Association - U.S. Western Lead

2023 - 2025

Oversaw 30+ chapters in California, Oregon, Washington, Nevada, Idaho, and Arizona. Coordinated events, created educational labs, and supported chapter growth. Led Glendora High (USA011) to become the top chapter nationwide.

Additional projects: Brain Computer Interface, Urban Heat Mitigation Planner, Gatcha Thrifting Machine, Bentopad (custom macropad), PCB Business Cards, Electrocardiogram. Full Portfolio: <https://www.changchang.me/portfolio.html>

Skills

Hardware: PCB Design (Kicad, EasyEDA), CAD (Fusion360, Onshape), Embedded Systems, ESP32/ESP-IDF, SPI/I2C/I2S, Power Management, SMT soldering | Programming: C++/C, Python, Java, R, HTML/CSS, JavaScript